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Elevate Labs Internship: Task 7

Objective: Use SQL inside Python to pull simple sales info (like total quantity sold, total revenue), and display it using basic print statements and a simple bar chart.

1. import sqlite3

2. Create SQLite database

```
conn = sqlite3.connect("sales_data.db")
cursor = conn.cursor()
```

3. Create a simple sales table

```
cursor.execute("""
    CREATE TABLE sales (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        product TEXT,
        quantity INTEGER,
        price REAL,
        category TEXT,
        date TEXT,
        region TEXT
    )
""")
```

```
In [4]: ► cursor.execute("""
        CREATE TABLE sales (
            id INTEGER PRIMARY KEY AUTOINCREMENT,
            product TEXT,
            quantity INTEGER,
            price REAL,
            category TEXT,
            date TEXT,
            region TEXT
        )
        """)
```

```
Out[4]: <sqlite3.Cursor at 0x23c25ec0b90>
```

4. Insert sample data

```
sales_data = [
    ("Apples", 10, 1.5, "Fruit", "2025-04-01", "North"),
    ("Bananas", 20, 0.8, "Fruit", "2025-04-01", "South"),
    ("Cherries", 15, 2.0, "Fruit", "2025-04-02", "East"),
    ("Chips", 30, 1.0, "Snack", "2025-04-03", "West"),
    ("Soda", 25, 1.2, "Drink", "2025-04-03", "North"),
    ("Bananas", 10, 0.8, "Fruit", "2025-04-04", "South"),
    ("Apples", 5, 1.5, "Fruit", "2025-04-05", "East")
]

cursor.executemany("""
    INSERT INTO sales (product, quantity, price, category, date, region)
    VALUES (?, ?, ?, ?, ?, ?)
""", sales_data)
```

```
In [5]: sales_data = [
        ("Apples", 10, 1.5, "Fruit", "2025-04-01", "North"),
        ("Bananas", 20, 0.8, "Fruit", "2025-04-01", "South"),
        ("Cherries", 15, 2.0, "Fruit", "2025-04-02", "East"),
        ("Chips", 30, 1.0, "Snack", "2025-04-03", "West"),
        ("Soda", 25, 1.2, "Drink", "2025-04-03", "North"),
        ("Bananas", 10, 0.8, "Fruit", "2025-04-04", "South"),
        ("Apples", 5, 1.5, "Fruit", "2025-04-05", "East")
    ]

In [6]: cursor.executemany("""
        INSERT INTO sales (product, quantity, price, category, date, region)
        VALUES (?, ?, ?, ?, ?, ?)
        """, sales_data)
```

```
Out[6]: <sqlite3.Cursor at 0x23c25ec0b90>
```

```
conn.commit()
conn.close()
```

5. Reconnect to the database

```
conn = sqlite3.connect("sales_data.db")
```

6. Query to show the actual full table

```
query_all = "SELECT * FROM sales"
```

7. Load into DataFrame and print table

```
df_all = pd.read_sql_query(query_all, conn)
print("Full Sales Table:")
print(df_all)
```

```
In [11]: ► df_all = pd.read_sql_query(query_all, conn)
print("Full Sales Table:")
print(df_all)
```

```
Full Sales Table:
   id  product  quantity  price  category  date  region
0   1   Apples        10    1.5    Fruit  2025-04-01  North
1   2  Bananas        20    0.8    Fruit  2025-04-01  South
2   3  Cherries        15    2.0    Fruit  2025-04-02   East
3   4   Chips        30    1.0    Snack  2025-04-03   West
4   5    Soda        25    1.2    Drink  2025-04-03  North
5   6  Bananas        10    0.8    Fruit  2025-04-04  South
6   7   Apples         5    1.5    Fruit  2025-04-05   East
```

8. Query to summarize total quantity and revenue per product

```
query_summary = """
SELECT product,
       SUM(quantity) AS total_qty,
       SUM(quantity * price) AS revenue
FROM sales
GROUP BY product
"""
```

9. Load summary into DataFrame

```
df_summary = pd.read_sql_query(query_summary, conn)
print("\nSales Summary:")
print(df_summary)
```

```
In [13]: ► df_summary = pd.read_sql_query(query_summary, conn)
print("\nSales Summary:")
print(df_summary)
```

```
Sales Summary:
   product  total_qty  revenue
0   Apples         15    22.5
1  Bananas         30    24.0
2  Cherries         15    30.0
3   Chips         30    30.0
4    Soda         25    30.0
```

```
conn.close()
```

10. Create bar chart

```
import matplotlib.pyplot as plt
```

```
df_summary.plot(kind='bar', x='product', y='revenue', legend=False,  
color='lightcoral')  
plt.ylabel("Revenue")  
plt.title("Revenue by Product")  
plt.tight_layout()  
plt.savefig("sales_chart.png") # Saves the chart image  
plt.show()
```

```
df_summary.plot(kind='bar', x='product', y='revenue', legend=False, color='lightcoral')  
plt.ylabel("Revenue")  
plt.title("Revenue by Product")  
plt.tight_layout()  
plt.savefig("sales_chart.png") # Saves the chart image  
plt.show()
```

